Comparative Study of Quantum Programming Frameworks Using Grover's Algorithm

Students will work in groups of three. Each group must implement Grover's algorithm using Qiskit as the mandatory reference environment. In addition, each group will choose two environments among Cirq, Amazon Braket, and PennyLane to implement Grover's algorithm. The deliverable will therefore consist of three codes for the three environments.

The objective of this lab is to compare how different quantum programming environments support quantum circuit construction, oracle implementation, execution, and measurement. Students will implement Grover's search algorithm, execute the corresponding quantum circuits, and analyze the evolution of the probability amplitude of the marked state.

The following questions should be addressed during the lab:

1. How is the uniform superposition state generated from the initial state $|00\ldots0\rangle$?
2. How is the oracle implemented to mark the target state $|\omega\rangle$?
3. How is the diffusion operator Dimplemented in each environment?
4. How does the probability distribution evolve after each Grover iteration?
5. Demonstrate why the optimal number of Grover iterations is approximately
$$k \approx \frac{\pi}{4}\sqrt{N}$$
and explain why applying additional iterations can decrease the probability amplitude of the marked state $|\omega\rangle$.
6. What are the main differences between Qiskit and the two selected environments in terms of syntax, abstraction level, usability, and execution workflow?

The session of 27th May will be mainly devoted to lab work. **All students are required to attend. Any unexcused absence will result in penalties.**

The deliverable will be in the form of a document containing a link to the Git repository for the developed code. It must be submitted before Monday, June 1, 6:00 p.m.

Group interviews (30 minutes, via Teams) will be scheduled during the week of June 1 – 5 to evaluate the lab. The three students of each group must be present for this Teams session. Students must prepare a presentation for this purpose. It is not part of the deliverable due by Monday, June 1.